



Identification of multiple rare earths and associated elements in raw monazite sands by laser-induced breakdown spectroscopy

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ABSTRACT

Laser-induced breakdown spectroscopy (LIBS) is an emerging and promising tool for rapid and efficient elemental analysis. We used the LIBS technique to analyze the raw monazite sands collected from the beaches in southern Bangladesh. We have detected a number of rare earths (lanthanides) such as cerium, lanthanum, praseodymium, neodymium, yttrium, ytterbium, gadolinium, dysprosium and erbium, in addition to other metallic elements such as zirconium, chromium, titanium, magnesium, manganese, niobium and aluminum. To the best of our knowledge, this is the first time multiple rare earths have been identified in a natural mineral by LIBS.

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1. Introduction

Laser-induced breakdown spectroscopy is a rapidly evolving and versatile technique for rapid and efficient elemental analysis [1–3]. In this technique, a high-power pulsed laser is focused on the sample to generate a high-temperature transient plasma, and a high-resolution spectrometer in conjunction with a gated ICCD is used to analyze the light emitted from it. This technique has found many applications in diverse fields, such as industry [4,5], environmental monitoring [6], mars exploration [7], coal and mineral analysis [8–10], etc.

Beach sands in a number of countries around the world such as India, China, Australia, Brazil and Bangladesh contain heavy minerals such as magnetite, monazite, ilmenite, rutile and zircon in addition to other minor minerals [11]. Monazite, zircon and rutile are commercially important minerals from which rare earths, zirconium and titanium (and their compounds) are extracted in a number of countries around the world. Of these minerals, probably the most chemically complex mineral is monazite, which contains a number of rare earths (Lanthanides) in varying proportions. Since the rare earths have very similar chemical properties, identification and separation of these natural lanthanides are not straightforward. Usually, liquid or ion chromatography is used for their chemical identification [12], and complicated fractional crystallization or solvent extraction processes are required for their chemical separation [11,13].

To date, a few papers have been published about the analysis of rare earths by laser-induced breakdown spectroscopy (LIBS) [14–16]. Most of these publications are concerned with the detection of a few, known elements in lab-prepared samples of lanthanide compounds. To the best of our knowledge, there are no published reports concerned with the detection of rare earths in a naturally occurring mineral by LIBS. Since natural minerals like monazite sands contains other elements (such as silicon, titanium and iron) having their own, strong emission lines that may interfere with the emission lines of rare earths, the detection and identification of rare earths in these naturally occurring samples are not a trivial task.

In this paper, by using the LIBS technique, we have analyzed raw monazite sands from the monazite-rich beach sands from Bangladesh, and identified a number of rare earths in presence of strongly interfering elements such as iron and titanium. To our knowledge, this is the first time where multiple rare earths have been identified in a naturally occurring geological sample.

2. Experimental setup

The schematic diagram of the experimental LIBS system is shown in Fig. 1. The excitation laser is a Q-switched Nd:YAG laser system (Spectra-Physics LAB-170-10) that has output at fundamental wavelength of 1064 nm with pulse duration of 8 ns, repetition rate of 10 Hz and a maximum pulse energy of 850 mJ. The laser is also equipped with harmonic generators capable of generating the second, third and fourth harmonics of the fundamental by means of KDP crystals. The maximum pulse energy at the second harmonic (532 nm) is 450 mJ, but the actual

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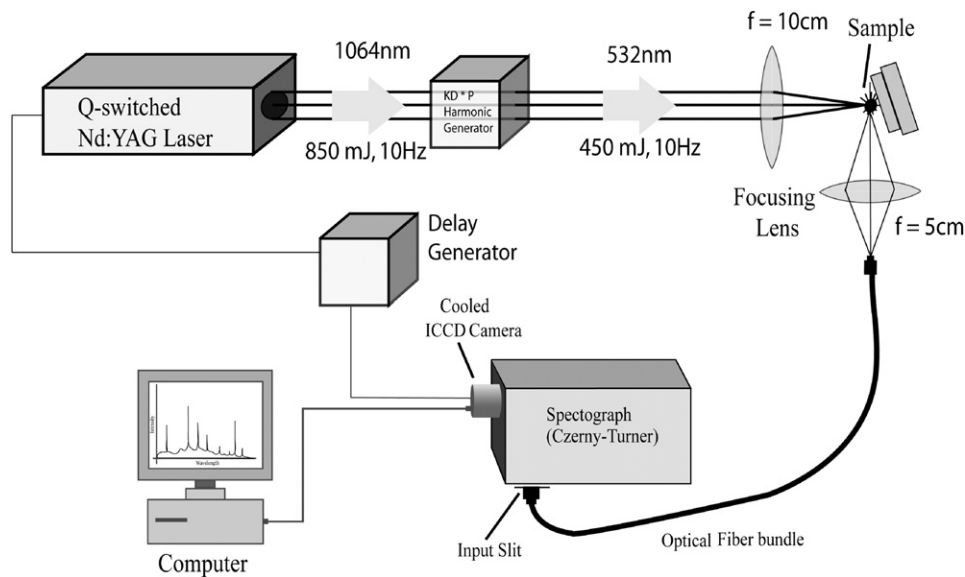


Fig. 1. Schematic diagram of the LIBS experiment.

pulse energy used in the experiments is about 35 mJ. The beam has a Gaussian profile in the far-field and has a beam divergence of less than 0.5 mrad.

Intense, transient plasma is produced by focusing the laser beam on the sample by a convex lens of 100 mm focal length. The spot size at the sample position is about 200 μm and the peak power density for 35 mJ pulses is of the order of 14 GW/cm^2 . The plasma is generated at the repetition rate of the laser, i.e. at 10 times per second. The light emitted from the plasma is focused by a fused quartz lens ($f=50\text{ mm}$) and collected by a 3 m long multimode silica optical fiber. The light is then transmitted through the fiber to its other end, which is placed at the entrance slit of a 750 mm focal length computerized Czerny-Turner spectrograph (Acton Model SP-2758). The spectrograph is equipped with three ruled gratings: 2400, 600 and 300 grooves/mm blazed at 240, 500 and 300 nm, respectively, which are interchangeable under computer control, providing high- and low-resolution spectra in the wavelength range 190–850 nm. If 600 grooves/mm grating is used, a spectrum of about 38 nm width can be captured without moving the grating, and for the 2400 grooves/mm grating, it is about 9 nm. However, this width (9 nm) is insufficient to cover all the wavelengths of the emission spectrum from the sample. Hence, to cover a wider spectral range, the grating is stepped under computer control, and a separate spectrum is acquired for each grating position.

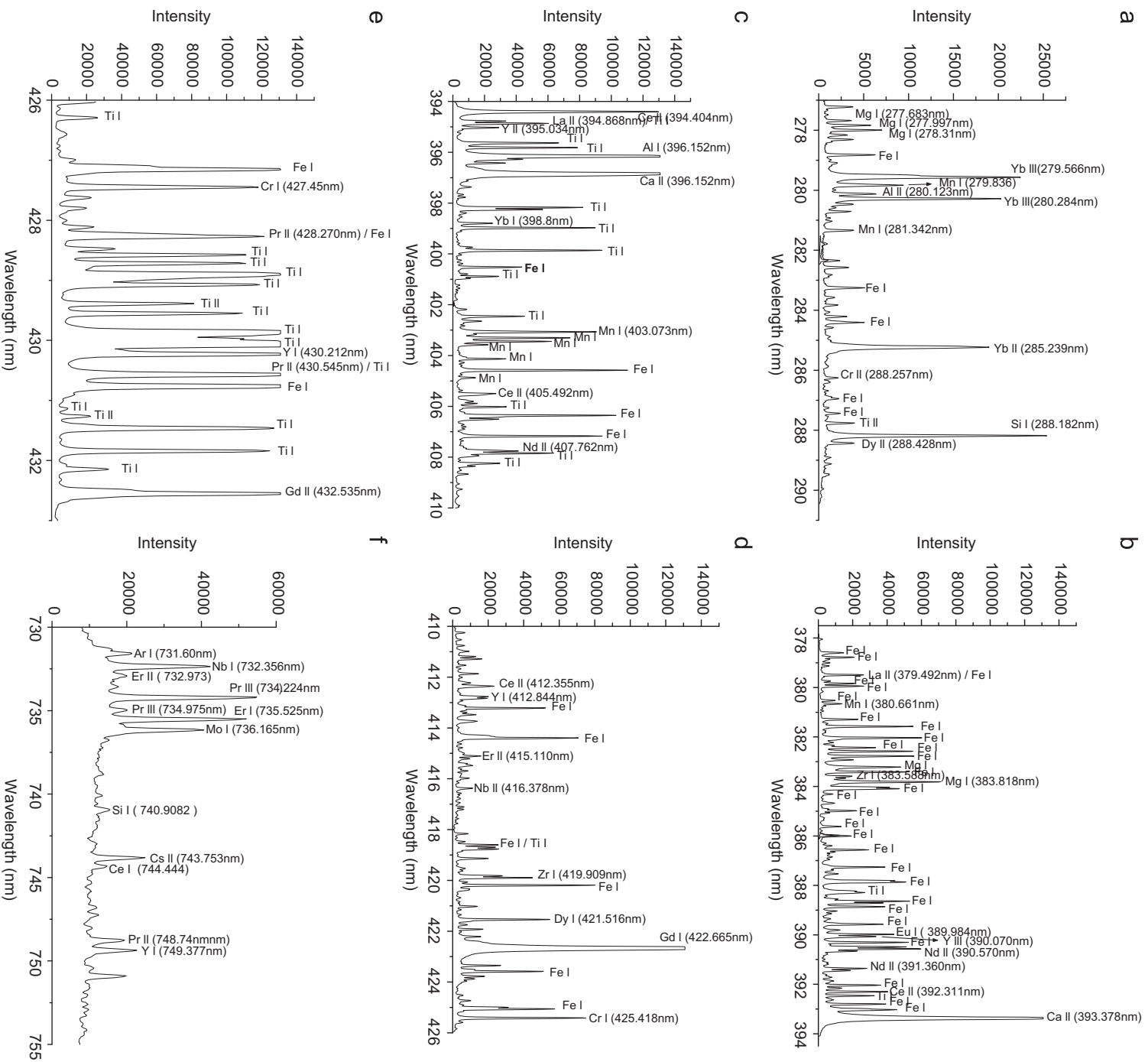
The output end of the spectrograph is coupled with an intensified and gated CCD camera (Princeton PI-MAX with Unigen II coating and programmable delay generator). The ICCD camera has 1024×1024 pixels and is cooled to -20°C by a Peltier cooler to reduce noise. The ICCD camera is electrically triggered by the Nd:YAG laser pulses after a software-controlled, adjustable time delay. In this way, the intense background initially created by the high-temperature plasma can be largely eliminated, and the atomic emission lines of the elements can be more clearly observed. In the present experiments, a delay time of 1.5 μs and a gate width of 50 μs are selected for the optimum signal. Usually, spectra from a number of laser shots (about 40–80) are acquired and averaged to increase the signal-to-noise ratio. The spectrum captured by the CCD camera is transferred to the personal computer by USB cable. All the functions of the ICCD camera and the Acton spectrograph can be fully controlled by WinSpec/32 software provided by the manufacturer.

3. Sample processing

The sands collected from the beach are first washed and dried. They are panned in attempts to separate the lighter sands from the heavier and darker colored sands containing the heaviest mineral monazite. Even after this separation, the sand contains other constituents like rutile, ilmenite, etc. The dark colored sands containing the monazite are ground and passed through 75 μm sieve and finally, small pellets of monazite-enriched sand sample are prepared by using a hand press with sufficient pressure (~ 80 bars). About 10 pellets are made and subjected to LIBS experiments as described above.

4. Results

The acquired LIBS spectra of a typical sample collected from the beaches in Cox's Bazaar area, Chittagong, are shown in Fig. 2(a)–(f) for some selected spectral regions. About 40–80 shots of the laser are averaged to acquire each spectrum. The wavelengths of the emission lines are independent of the energy of the incident pulses, but the intensity of the lines increases with the pulse energy up to the saturation limit. The average plasma temperature during the observation time window, estimated by the standard Boltzmann plot method [1], is in the range of 9000–11000 K, and will be reported elsewhere. Each of the observed line peaks in the spectra is compared with the online LIBS database available from the US National Institute of Standards and Technology [17]. Principal identified lines are shown in the figures. A large number of strong lines of titanium and iron are detected and these sometimes cause severe interference with the expected emission lines of lanthanides. A lanthanide is considered to be detected in our experiments only when at least three of its non-interfered lines are unambiguously identified. The principal identified lanthanides with their ionic states, the observed values of the wavelengths of the detected emission lines and total number of unambiguously identified lines are shown in Table 1, along with the possible interfering lines from other elements. Table 2 also shows the lines of other elements, both metallic and non-metallic, co-existing with the monazite sands. Only non-interfered, unambiguously identified lines are shown. A significant number of transition elements can



5. Discussions

be detected, whose origin will be explained later. In addition, both silicon and phosphorus can be identified as the major non-metals.

In the above experiments, a large number of elements are simultaneously detected in the raw monazite sands. Most interesting among the elements are the lanthanides, such as

cerium, lanthanum, praseodymium, neodymium, yttrium, ytterbium, gadolinium, dysprosium and erbium. It is a well-known fact that depending on the origin of the rocks from which they are formed, the monazite sands around the world show some significant variations in their lanthanide compositions [11]. For example, Indian monazite sands, which have been extensively studied by physical and chemical methods, contain principally mixtures of cerium (Ce), lanthanum (La), neodymium (Nd), praseodymium (Pr) and samarium (Sm) phosphates, along with

Table 1
Emission lines of identified lanthanides.

Elements identified	Wavelengths of major identified lines with ionic states (nm) [*strong lines]	Possible interfering lines from other elements (nm)	Total number of identified lines
Cerium (Ce)	305.728(III)*, 308.377(II), 349.757(III)*, 350.459(III)*, 362.349(II), 389.806(II), 392.311(II), 394.404(II)*, 405.492(II), 412.355(II)*, 514.757(II), 744.444(I) 794.842(III)	None	13
Lanthanum (La)	375.909(II), 379.131(II), 379.492(II), 394.868(II), 403.148(II), 408.670(II)*, 415.979(I)*, 423.876(II)*, 426.983(II)*, 492.170(II)	Ti(I) 375.930, Fe(I) 379.5002 Ti(I) 492.177	7
Neodymium (Nd)	348.494(I), 352.765(II), 390.570(II)*, 391.360(II), 407.762(II), 495.477(I)*, 510.772(II)*, 527.334(II), 629.755(I)	Ce(II) 348.505	8
Praseodymium (Pr)	346.556(II)*, 428.270(II)*, 430.545(II)*, 440.841(II)*, 445.454(II), 615.984(III)*, 670.661(III)*, 734.224(III)*, 734.975(III), 786.653(III), 788.738(III)	Fe(I)428.24 nm, Ti(I)430.592 nm, Sm(II) 445.463 nm	8
Yttrium (Y)	228.432(III)*, 291.900(I), 332.799(II)*, 378.852(II), 390.070(III)*, 395.034(II)*, 410.251(I)*, 412.844(I)*, 430.212(I), 485.984(I)*, 705.347(I), 749.377(I)	None	12
Ytterbium (Yb)	232.031(II), 266.680(III), 279.566(III)*, 280.284(III)*, 285.239(II)*, 398.800(I)	None	6
Gadolinium (Gd)	229.242(III)*, 308.99(II), 345.028(II), 346.864(II), 364.136(II), 364.622(II), 422.665(I)*, 432.535(II), 440.658(II), 509.883(II)*	Fe(I) 229.2524	9
Dysprosium (Dy)	288.428(II), 317.803(II), 347.387(II), 349.87(II)*, 352.392(II)*, 353.168(II)*, 353.509(II)*, 353.621(II)*, 358.508(II), 421.516(I), 432.535(II), 634.331(I)*	None	12
Erbium (Er)	226.889(III)*, 235.885(II)*, 323.198(II), 341.778(II), 414.292(II), 415.110(I)*, 512.745(II)*, 649.205(I)*, 732.973(II), 735.525(I), 793.656(I)	Y(I) 414.285, Fe(I) 793.7131	9
Samarium (Sm)	327.281(II)*, 415.211(II)*, 445.454(II)*, 484.408(II)*	Pr(II) 445.468 nm	3
Europium (Eu)	321.059(I), 352.098(II)*, 389.984(I)*, 443.546(I)	Ca(I) 443.569	3

Table 2
Other detected elements.

Elements identified	Wavelengths of major identified lines with ionic states (nm)	Total number of identified lines
Magnesium (Mg)	189.664(III), 277.683(I), 277.997(I), 278.310(I), 383.218(I), 383.818(I), 517.279(I), 518.361(I)	8
Chromium (Cr)	223.350(III), 359.339(I), 425.418(I), 427.450(I), 520.450(I), 520.604(I), 520.841(I), 534.596(I), 534.849(I)	9
Calcium (Ca)	317.925(II), 364.442(I), 393.378(II), 396.152(II), 443.479(I) 443.546(I), 458.597(I), 534.964(I), 559.799(I), 854.017(II), 866.053(II)	11
Aluminum (Al)	266.043(I), 280.123(II), 308.222(I), 309.280(I), 396.152(I)	5
Zirconium (Zr)	300.38(II), 318.276(II), 322.259(II), 328.45(II), 344.728(I), 383.588(I), 419.909(I), 485.147(I), 488.123(I), 708.673(I), 709.669(I), 711.086(I), 716.827(I)	13
Manganese (Mn)	279.836(I), 281.342(I), 293.300(II), 293.920(II), 294.908(II), 304.06(I), 380.661(I), 403.073(I), 403.307(I), 403.449(I), 403.578(I), 404.135(I), 404.875(I), 413.479(I), 475.397(I), 476.228(I), 476.630(I), 482.362(I)	18
Niobium (Nb)	231.307(III), 347.852(I), 416.378(I), 548.102(I), 654.460(I), 732.356(I)	6
Tantalum (Ta)	248.397(II), 289.900(I), 644.406(I)	3
Copper (Cu)	211.249(II), 222.62(II), 223.001(I), 226.094(I), 226.307(I), 229.416(I), 695.233(II), 766.310(II)	8
Zinc (Zn)	198.543(II), 202.559(II), 220.96(II), 761.286(II), 779.910(I)	5
Silicon (Si)	243.492(I), 250.694(I), 251.437(I), 251.615(I), 251.921(I), 252.419(I), 252.855(I), 263.133(I), 288.182(I), 740.9082(I)	10
Phosphorus (P)	253.571(II), 255.316(I), 422.214(III), 446.381(III), 507.935(I), 510.961(I), 646.083(II), 710.222(I)	8
Titanium (Ti)	323.427(II), 323.632(II), 323.874(II), 324.175(II), 324.834(II), 325.164(II), 325.266(II), 325.401(II), 399.866(I), 451.252(I), 451.780(I), 452.255(I), 452.707(I), 453.301(I), 453.453(I), 453.554(I), 454.444(I), 454.852(I), 455.222(I), 455.524(I), 499.101(I), 499.944(I), 500.713(I), 501.417(I), 501.609(I), 501.994(I), 502.476(I), 503.553(I), 503.574(I), 503.641(I), 503.831(I), 503.983(I)...	
Iron (Fe) (partial list)	259.845(II), 259.949(II), 355.841(I), 356.530(I), 357.006(I), 381.289(I), 381.577(I), 382.035(I), 382.434(I), 382.579(I), 382.778(I), 384.030(I), 384.100(I), 384.987(I), 404.581(I), 406.357(I), 526.965(I), 527.044(I)...	

minor quantities of other lanthanides along with other elements. In our LIBS experiments, all of the above elements were positively identified, along with minor lanthanides e.g. gadolinium (Gd), dysprosium (Dy), ytterbium (Yb), yttrium (Y), erbium (Er) and europium (Eu). All these show the unique multi-elemental capability of the LIBS technique, a capability not shared by many chemical identification methods. To the best of our knowledge, the detection of the elements praseodymium, ytterbium and dysprosium by LIBS in the present experiment is the first reported in the literature. These elements are probably present in minor or trace amounts and have hitherto remained undetected. Because of the relatively high detection limits of the LIBS technique (typically of the order of parts per million or lower), they were detected in the present experiments.

The detected non-metals in Table 2 are phosphorus and silicon, which is hardly surprising. Other metallic elements such as

niobium, tantalum, chromium, zirconium, manganese, calcium, copper, zinc, aluminum, titanium and iron were also detected with a high degree of certainty. Most of these are transition metals, and some of them (e.g. titanium, iron, magnesium and some other elements) have already been detected in the beach sands from the same geographical area by our group [18], using a non-gated, non-intensified LIBS system of fairly low sensitivity. In the present experiments, the vastly superior sensitivity afforded by the gated and intensified ICCD system probably enables us to detect a large number of transition elements, in addition to others.

The beach sands in many parts of the world, including India and Bangladesh, are known to contain minerals such as rutile, ilmenite, zircon, leucosene, sillimanite, magnetite and garnet in addition to monazites [11,19,20]. The detected elements titanium and iron in our experiments may originate from rutile, magnetite

and ilmenite, which are mainly, TiO_2 , Fe_2O_3 and FeTiO_3 , respectively. Ilmenite contains appreciable quantities of magnesium and manganese, accounting for the presence of magnesium and manganese lines in our experiments. Zircon is chiefly zirconium silicate, which accounts for the presence of zirconium lines. Sillimanite is a form of aluminum silicate, which accounts for the presence of the observed aluminum lines. Niobium and tantalum may have been contributed by columbite, which is sometimes found in placer deposits in India [11]. Finally, the calcium and chromium lines may have originated from some types of garnet (i.e. calcium garnets), which are known to be present in beach sands [11]. Copper and zinc may be present as impurities in garnet or other minerals. Some of the identified elements may be present in fairly small amounts, down to trace concentrations. Due to the extremely high sensitivity of the LIBS system, all of these elements were detected simultaneously, along with the large number of lanthanides from the monazite mineral.

6. Conclusion

We have simultaneously detected a number of rare earth elements such as cerium, lanthanum, praseodymium, neodymium, yttrium, samarium, gadolinium, dysprosium, ytterbium in addition to other associated elements in the beach sands of Bangladesh by the LIBS technique. To the best of our knowledge, this is the first time rare earth elements have been identified in monazite sands by LIBS, and also the first time, the elements praseodymium, ytterbium and dysprosium have been subjected to LIBS experiments. The next logical step is to perform a detailed quantitative analysis of the monazite sands by LIBS to determine the relative abundances of the rare earths (and other elements) in the monazite sands. Experiments are presently under way in our laboratory to perform such analysis.

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